BYoung Physical Therapy

Return to Sport Assessment

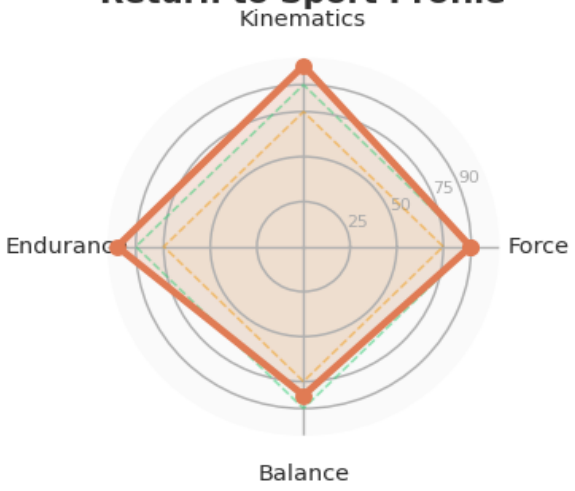
Patient: Claire Pulling **Surgical Side:** R **Date:** April 01, 2026 **Surgery:** 07/28/2025 (8.3 months post-op)

61

RTS Score
Caution
2 red flags · 8 mo post-op

Metric	Surgical	Non-Surg
DJ Peak GRF	676.8	766.1
DJ RFD 100ms	4548.1	6880.3
DJ Load Rate	6367.3	8172.7
DJ Land Sym	—	—
DJ Knee Flexion	106.4	105.7
DL Peak GRF	542.4	658.5

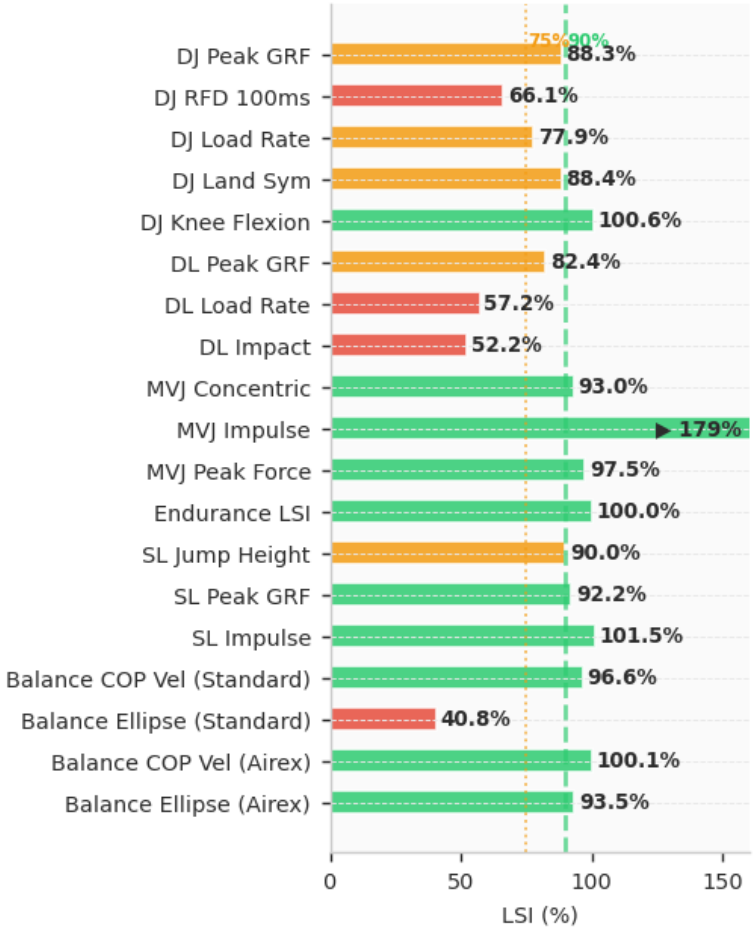
Return to Sport Profile



STRENGTHS	DETRACTORS

Perf 77 → ×0.86 red-flag → ×0.92 time → = 61/100

Limb Symmetry Index — All Tests



CLINICAL FLAGS

- DJ RFD 100ms: LSI 66.1% — significant asymmetry
- DL Load Rate: LSI 57.2% — significant asymmetry
- DL Impact: LSI 52.2% — significant asymmetry
- Balance Ellipse (Standard): LSI 40.8% — significant asymmetry

● ≥90% Symmetric ● ≥75% Progressing ● <75% Deficit LSI = Surgical ÷ Non-Surgical × 100

DROP JUMP

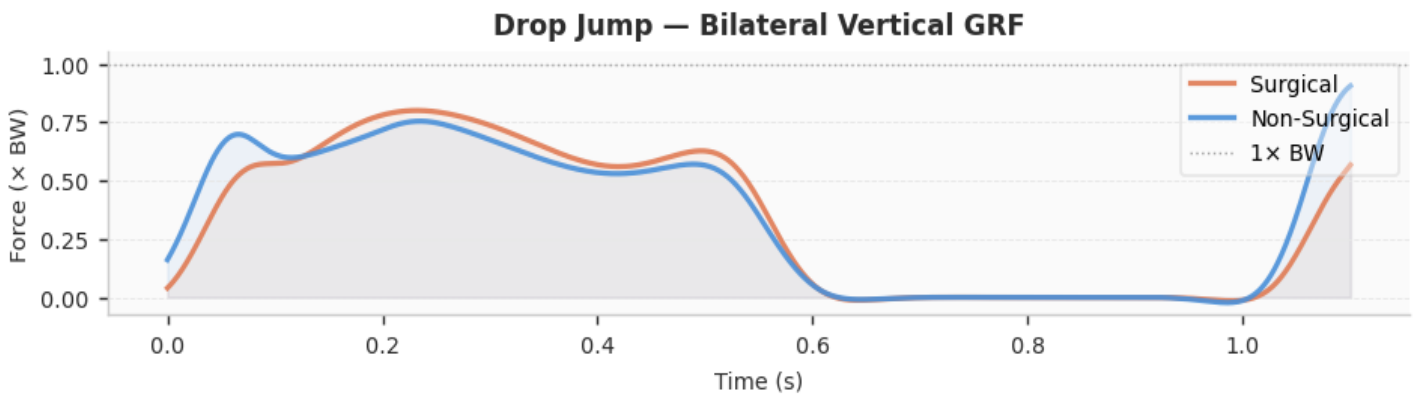
Metric	Surgical	Non-Surg	LSI
Peak GRF (N)	676.8	766.1	88.3%
Impulse (N·s)	310.2	319.6	97.0%
RFD 100ms (N/s)	4548.1	6880.3	66.1%
Loading Rate 50ms (N/s)	6367.3	8172.7	77.9%
Landing Sym 200ms (%) *(bilateral)*	—	—	88.4%
Impact Transient (N)	352.4	544.9	64.7%
Time-to-Peak (ms)	232.0	1100.0	—
RSI	1.137	—	—
Contact Time (s)	0.610	—	—
Peak Knee Flexion (°)	106.4	105.7	100.6%
Peak Knee Valgus (°)	11.0	8.3	132.3%
AP COP Mean — Surg / NS (mm)	1942.9	1925.6	—

CLINICAL INTERPRETATION

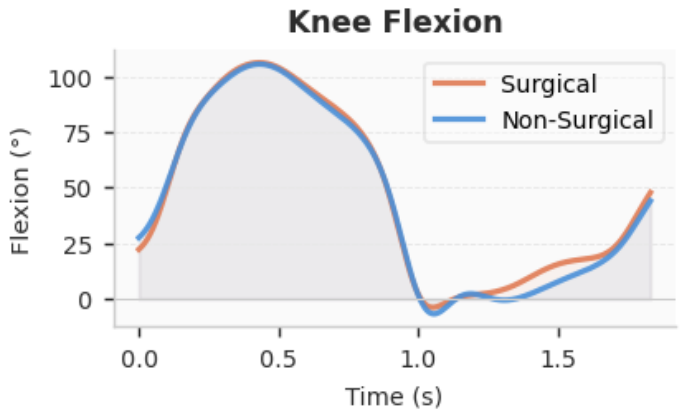
RSI 1.14 (■ below RTS target ≥ 1.30). Stiff-landing pattern likely — progressive plyometric loading indicated.

Landing symmetry 88% LSI: Surgical limb loading 12% less than non-surgical — surgical guarding detected.

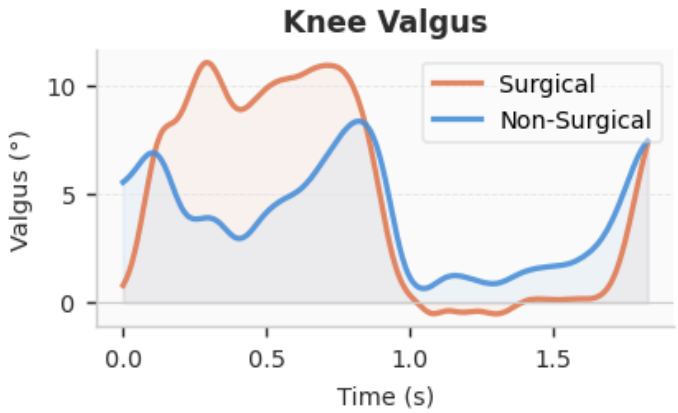
Targets: RSI ≥ 1.30 · Landing LSI 90–110% · Load rate LSI ≥ 90% · Knee flexion ≥ 60° · Valgus < 5°.



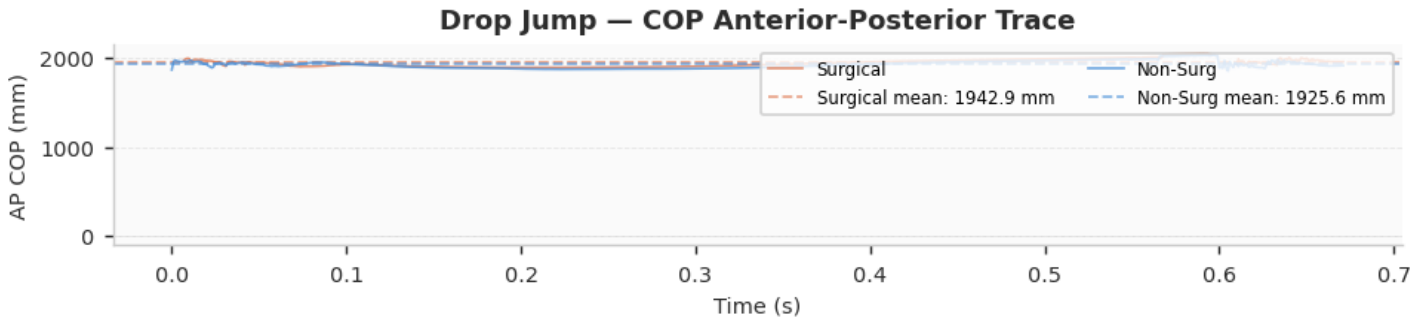
Bilateral GRF (× body weight). Orange = surgical, Blue = non-surgical.



Knee flexion. Target ≥ 60° at contact.



Knee valgus. Target < 5°. Elevated valgus = ACL re-injury risk.



AP COP trace during Drop Jump. Dashed lines = mean contact position per limb. Posterior shift on the surgical limb indicates guarding or reduced knee flexion at landing.

DROP LANDING

Metric	Surgical	Non-Surg	LSI
Peak GRF (N)	542.4	658.5	82.4%
Impulse (N·s)	300.7	311.3	96.6%
Loading Rate 50ms (N/s)	6016.0	10521.3	57.2%
Impact Transient (N)	293.0	560.7	52.2%
Peak Knee Flexion (°)	116.2	115.1	100.9%
Peak Knee Valgus (°)	12.5	7.3	170.2%
AP COP Mean — Surg / NS (mm)	1891.7	1898.1	—

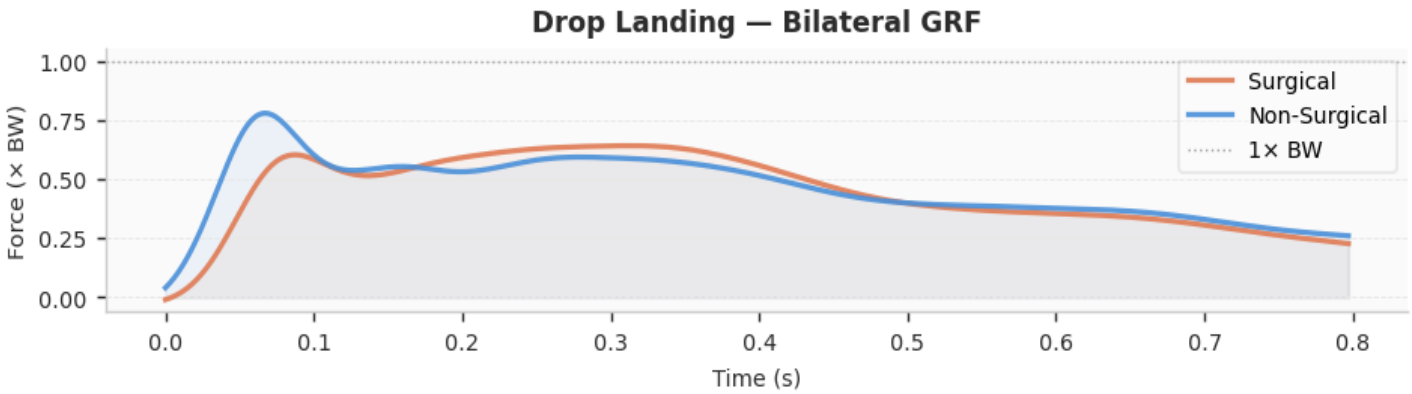
CLINICAL INTERPRETATION

Absorptive landing — assesses shock attenuation without a re-jump. Lower, more gradual force = better control.

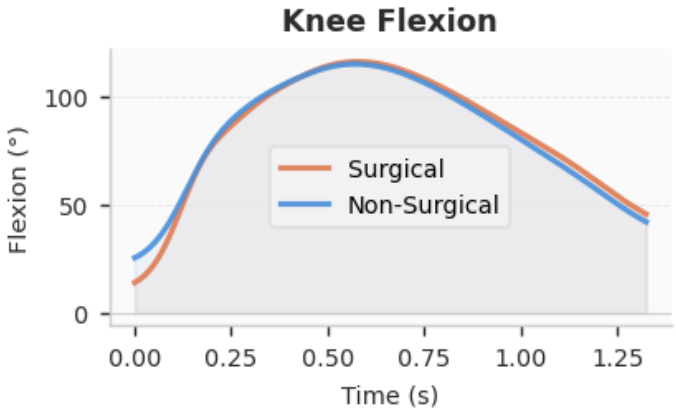
Load rate LSI 57%: Surgical limb loading rate 43% lower — surgical inhibition.

Impact transient LSI 52%. Target: < 2× BW peak, LSI 90–110%, no sharp spike.

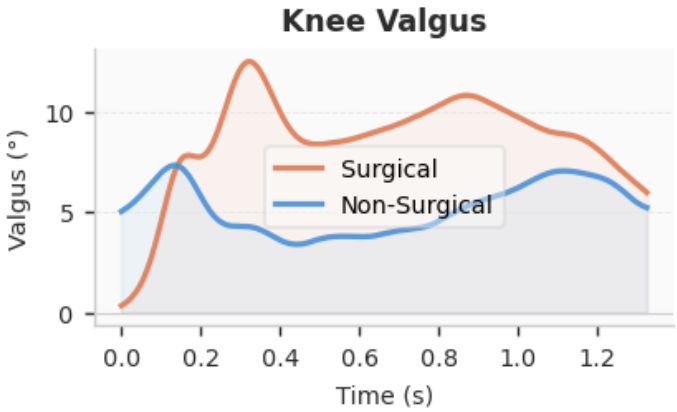
Targets: Peak GRF < 3× BW · Load rate LSI 90–110% · Knee flexion ≥ 60° · Valgus < 5°.



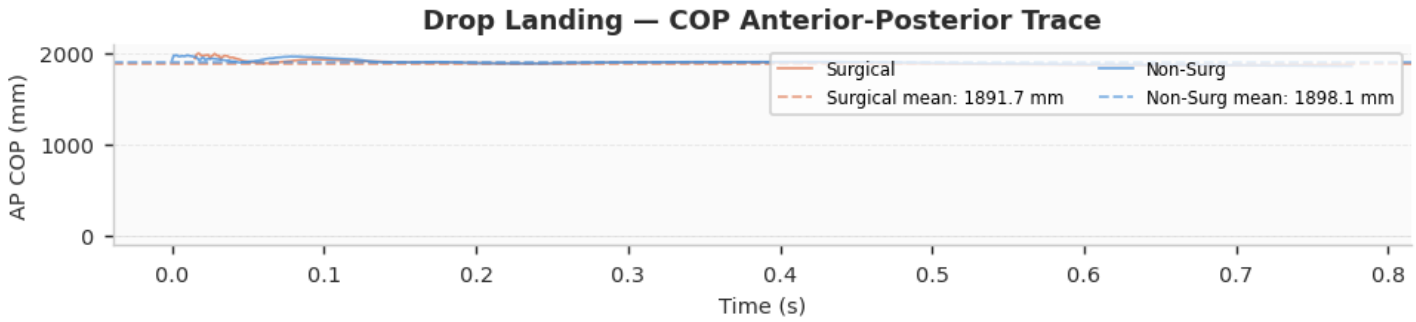
Bilateral GRF. Smooth progressive rise = good shock absorption. Sharp spike = stiff landing.



Knee flexion at landing.



Knee valgus at landing. > 8° surgical = elevated ACL stress.



AP COP trace during Drop Landing. A posterior shift on the surgical side indicates a heel-dominant strategy — the athlete is offloading the forefoot / reducing knee flexion depth.

MAX VERTICAL JUMP (CMJ)

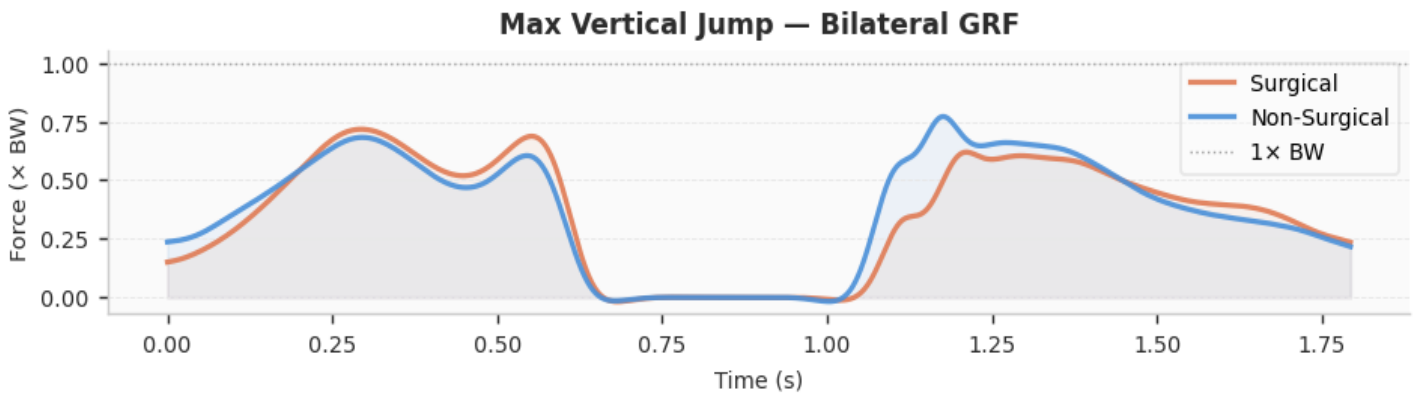
Metric	Surgical	Non-Surg	LSI
Peak Concentric GRF (N)	606.4	652.0	93.0%
Propulsion Impulse (N·s)	438.6	245.5	178.6%
Peak Force Surg / NS (N)	422.8	433.8	97.5%
Propulsion LSI (%) *(bilateral)*	—	—	178.6%
Jump Height	13.9	—	—
Flight Time (s)	0.381	—	—
Unweighting Imp Surg/NS	275.7	263.1	—
Peak Knee Flexion (°)	105.7	100.9	104.8%
Peak Knee Valgus (°)	11.3	8.8	128.1%
AP COP Mean — Surg / NS (mm)	2158.8	2157.0	—

CLINICAL INTERPRETATION

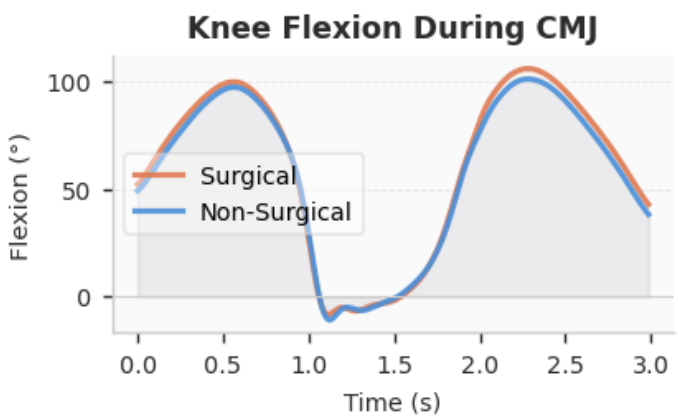
Bilateral CMJ — sensitive to subtle propulsion deficits masked at lower intensities.

Peak force LSI 97%: Symmetric bilateral output.

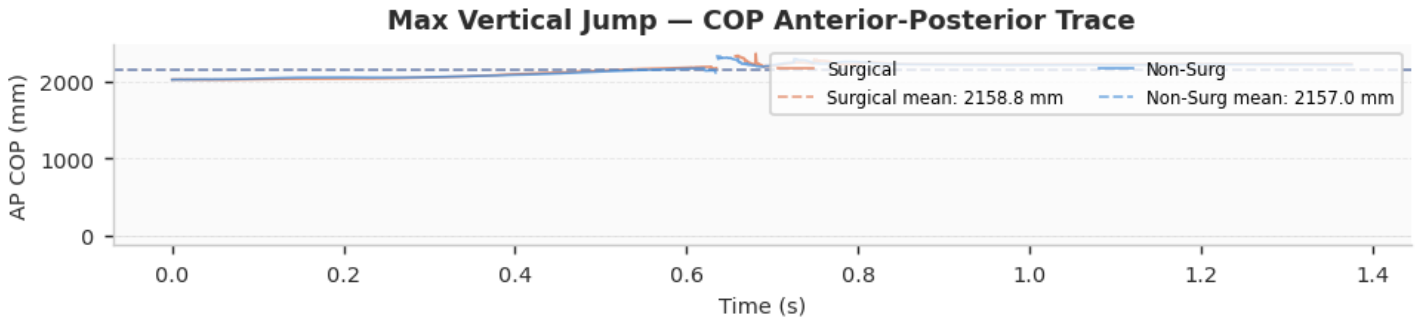
Propulsion LSI 179%. Targets: ≥ 90% propulsion, ≥ 90% peak force, matched unweighting impulse.



Full CMJ cycle: unweighting → eccentric → propulsion → flight. Propulsive peaks should be symmetric.



Knee flexion depth during CMJ. Asymmetric depth = compensatory strategy.



AP COP trace during CMJ. During propulsion, anterior COP shift drives jump height. Asymmetric anterior shift on the non-surgical side indicates the athlete is loading away from the surgical limb.

ENDURANCE SQUAT (30s)

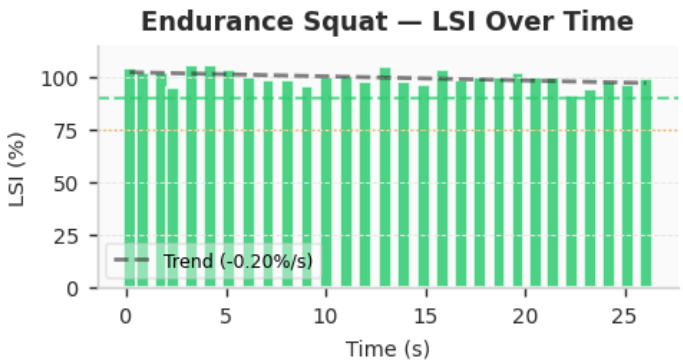
Metric	Surgical	Non-Surg	LSI
Mean Peak Force LSI (%) *(bilateral)*	—	—	**100.0%**
LSI First 10s (%) *(bilateral)*	—	—	**102.1%**
LSI Last 10s (%) *(bilateral)*	—	—	**98.2%**
Fatigue Drift (Δ%)	-3.9	—	—
Total Reps Detected	29.0	—	—
Mean Peak Force Surg (N)	557.4	—	—
Mean Peak Force NS (N)	557.8	—	—

CLINICAL INTERPRETATION

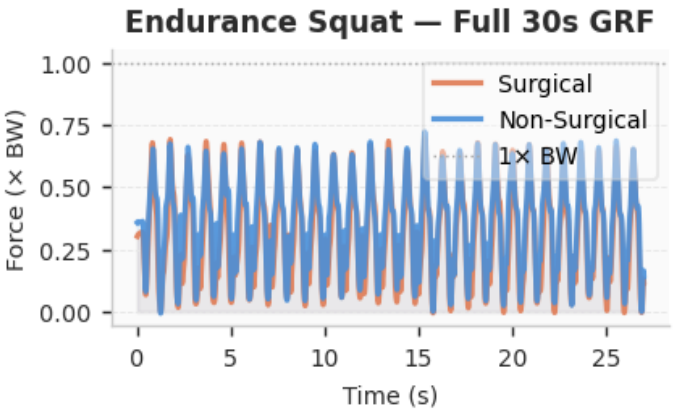
30s bilateral squat — reveals neuromuscular fatigue-induced asymmetry missed by single-rep testing.

Fatigue drift -3.9%: ✓ Symmetry maintained under fatigue.

Targets: Mean LSI ≥ 90% · Fatigue drift < 5% · No progressive offloading across 30s.



Per-rep LSI across 30s. Declining trend = neuromuscular fatigue breakdown.



Full 30s GRF trace. Force magnitude and rhythm across the set.

SINGLE-LEG VERTICAL JUMP

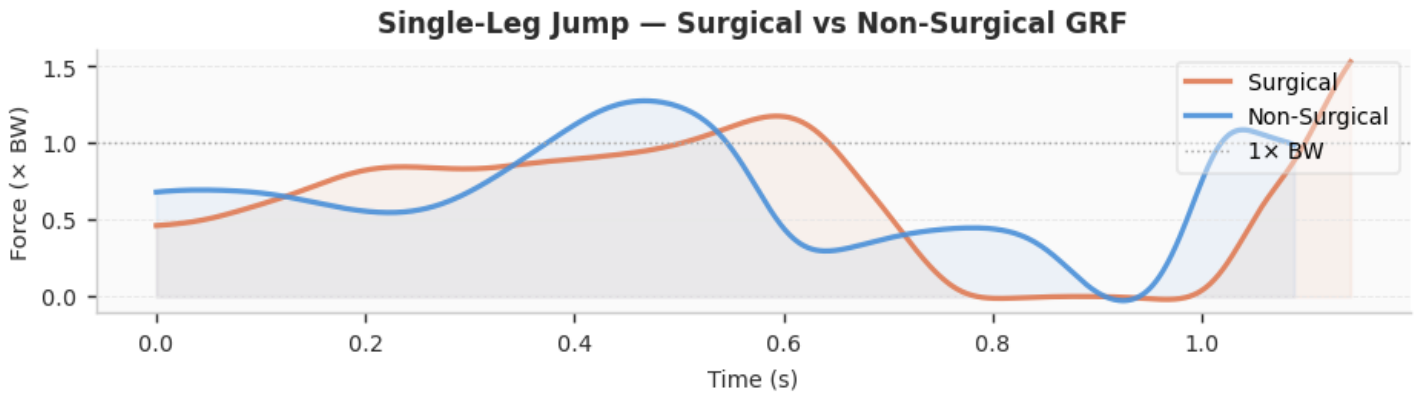
Metric	Surgical	Non-Surg	LSI
Jump Height	7.3	8.1	90.0%
Peak GRF (N)	995.1	1079.5	92.2%
Propulsion Impulse (N·s)	513.5	506.1	101.5%
RFD 100ms (N/s)	1165.7	295.1	—
Flight Time (s)	0.2	0.0	—
Contact Time (s)	0.8	0.9	—
Peak Knee Flexion (°)	61.3	55.0	—
Peak Knee Valgus (°)	11.3	10.7	—

CLINICAL INTERPRETATION

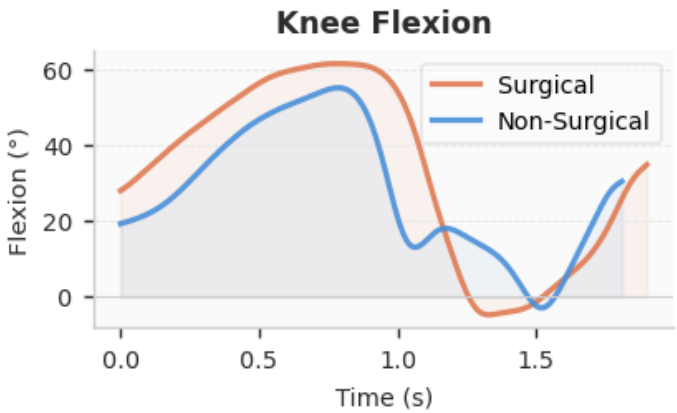
Highest-demand test — isolates each limb, removing bilateral compensation. Strongest re-injury risk predictor.

Jump height LSI 90%: ■ 10% below non-surgical. Continued plyometric loading required.

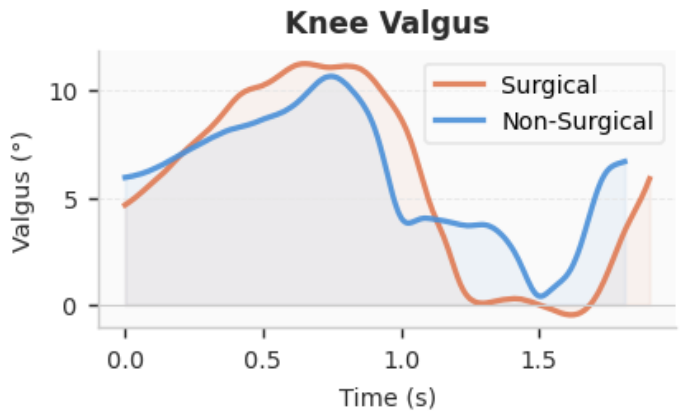
Targets: Jump height LSI ≥ 90% · Peak GRF LSI ≥ 90% · Valgus < 5° on both limbs.



Overlaid GRF for each limb tested separately. Propulsive peak height and shape should match.



Knee flexion strategy per limb.



Knee valgus per limb. Elevated surgical valgus under single-leg load = re-injury risk.

BALANCE — STANDARD (EYES CLOSED, FIRM SURFACE)

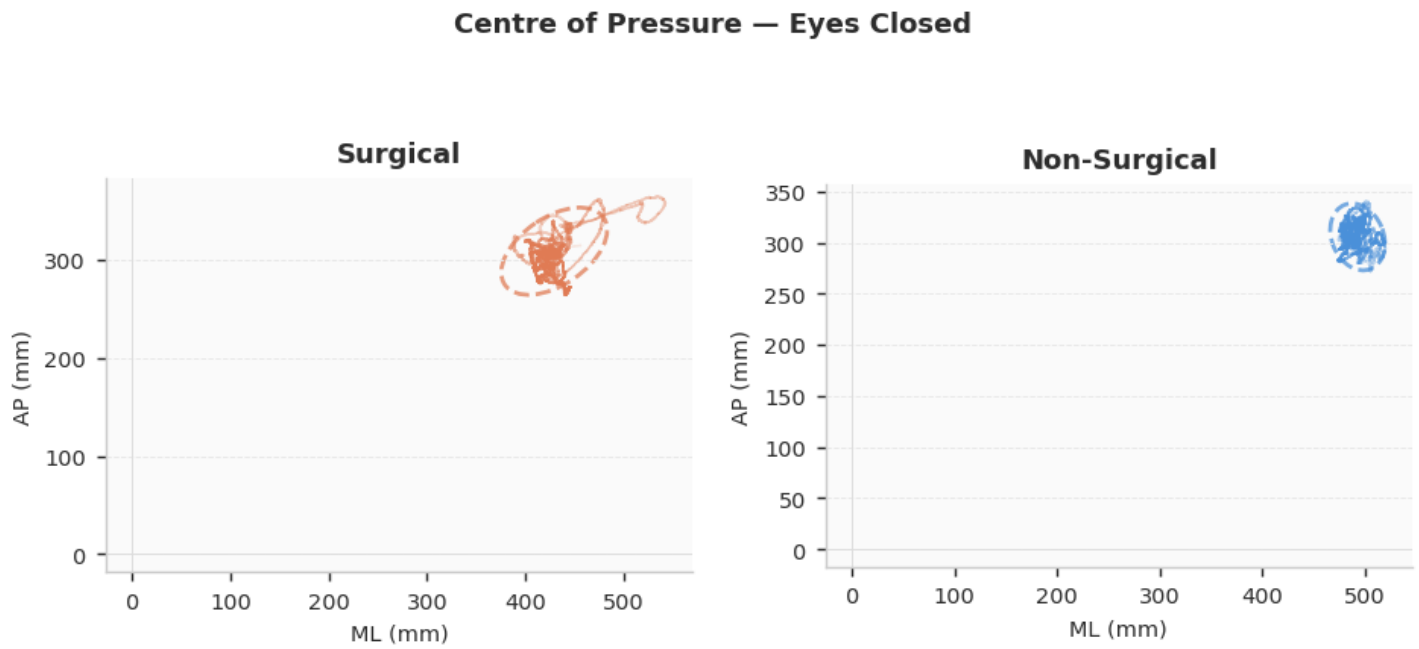
Metric	Surgical	Non-Surg	LSI
COP Mean Velocity (mm/s)	568.6	549.0	96.6%
COP Ellipse Area (mm²)	6568.0	2680.7	40.8%
Total Excursion (mm)	7959.1	7685.4	—
AP Range (mm)	100.8	68.7	—
ML Range (mm)	155.0	47.3	—
RMS Displacement (mm)	28.5	17.2	—

CLINICAL INTERPRETATION

Eyes-closed single-leg stance on firm surface. COP velocity is the most sensitive metric — higher speed = more corrective activity = reduced proprioception. Target: < 12 mm/s.

Surgical 569 mm/s vs Non-Surgical 549 mm/s — Surgical limb less stable. Proprioceptive retraining indicated.

COP Velocity LSI 97% (NS÷Surg×100 — lower = better). Normative target: < 12 mm/s. LSI target ≥ 90%.



COP trajectory — eyes closed. Tighter ellipse = better joint position sense. 95% confidence ellipse shown.

BALANCE — AIREX PAD (EYES CLOSED, UNSTABLE SURFACE)

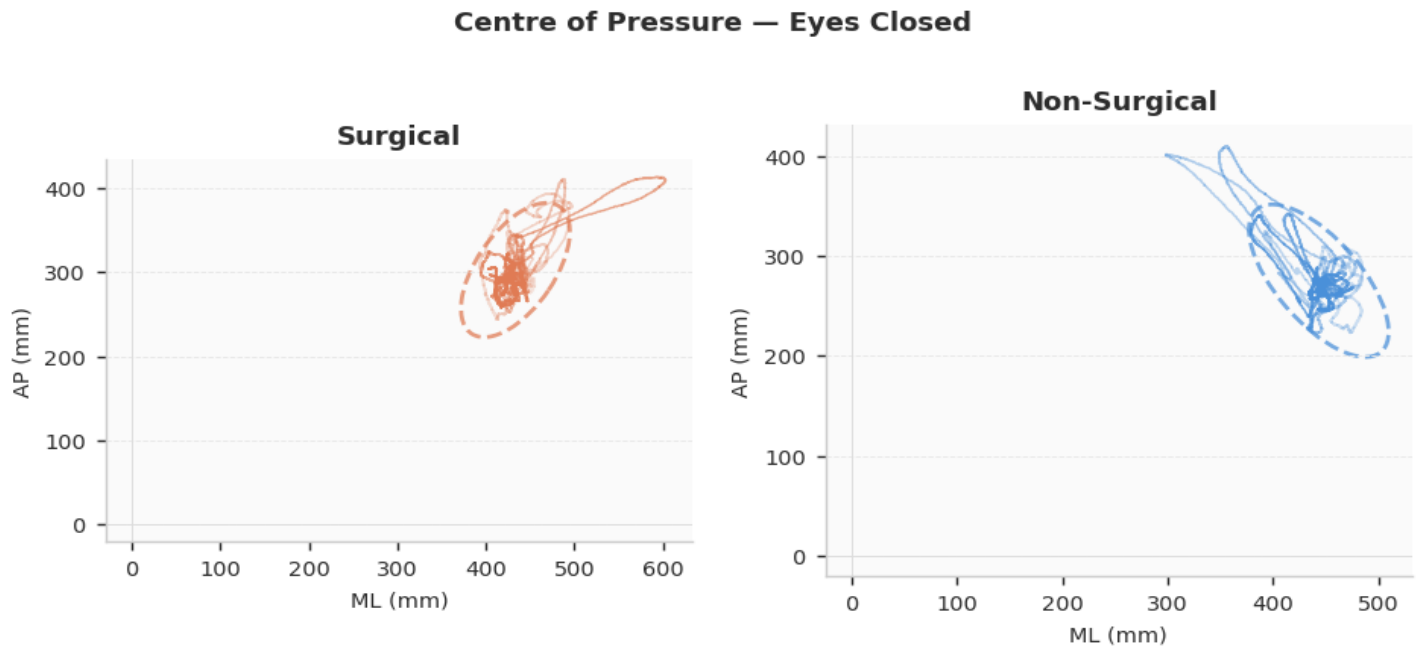
Metric	Surgical	Non-Surg	LSI
COP Mean Velocity (mm/s)	613.3	613.8	100.1%
COP Ellipse Area (mm²)	12650.5	11822.7	93.5%
Total Excursion (mm)	8586.1	8592.0	—
AP Range (mm)	169.9	188.1	—
ML Range (mm)	210.4	187.0	—
RMS Displacement (mm)	41.2	41.3	—

CLINICAL INTERPRETATION

Eyes-closed single-leg stance on Airex foam pad. Higher COP values are expected — the unstable surface challenges proprioceptive and neuromuscular systems. Symmetry between limbs (LSI ≥ 85%) is the key target.

Surgical 613 mm/s vs Non-Surgical 614 mm/s — COP velocity comparable between limbs.

COP Velocity LSI 100% (NS÷Surg×100 — lower = better). Normative target: < 50 mm/s. LSI target ≥ 90%.



COP trajectory — eyes closed. Tighter ellipse = better joint position sense. 95% confidence ellipse shown.